Additivity in Classical and Quantum Shannon Theory

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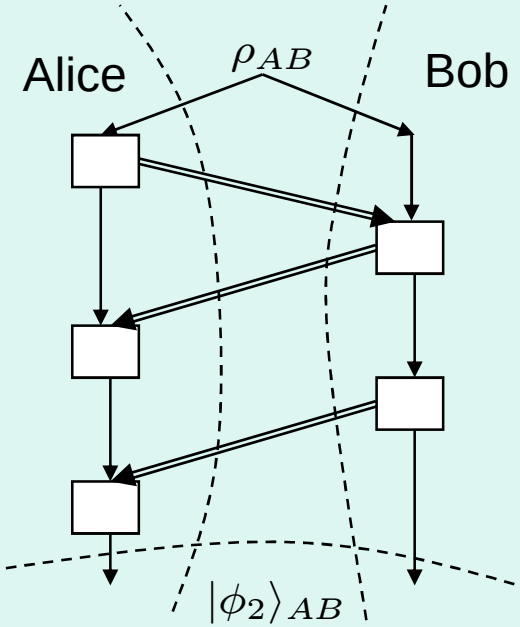
Joint work with

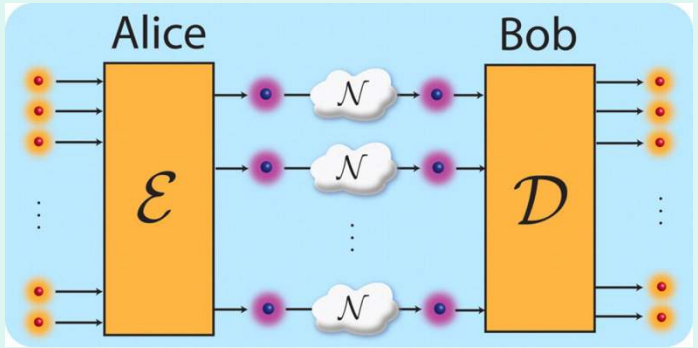
Andrew Cross and Ke Li

October 1, 2015

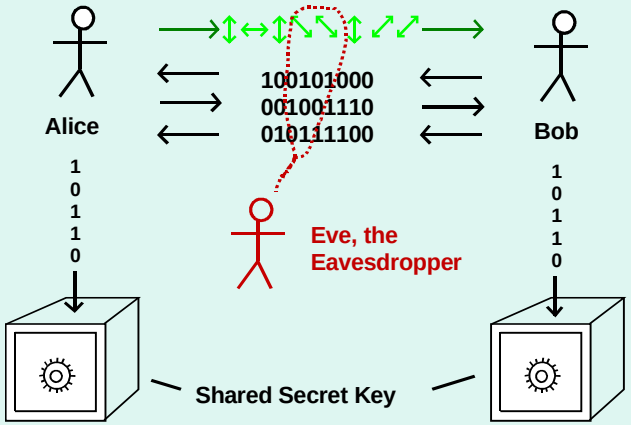
QCRYPT 2015, Tokyo, Japan

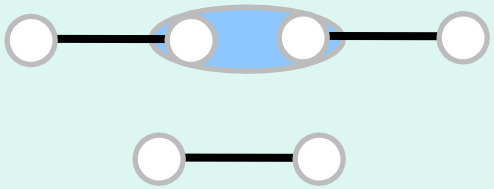
Information theory: sending, storing, processing data

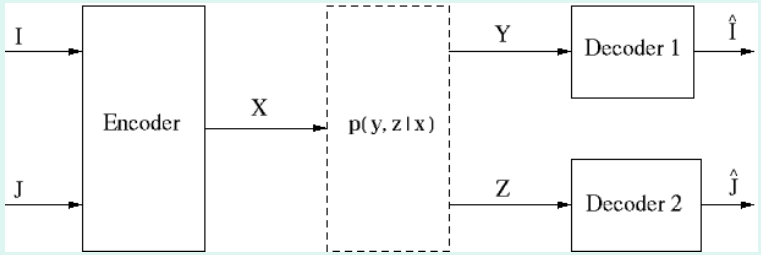




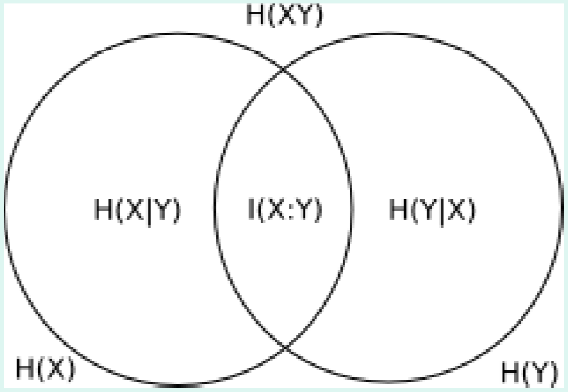








Entropies quantify the answers



- $H(X) = -\sum_x p_x \log p_x$
- $H(\rho) = -\mathrm{Tr}\ \rho \log \rho$

- Optimal Compression: H(X)
- Schumacher Compression: $H(\rho)$
- Classical Channel capacity: max I(X;Y)
  $I(X;Y) = H(X) + H(Y) - H(XY)$
- Quantum Communication: H(B) – H(E)
- Private capacity: I(V;B)-I(V;E)
- Strong Subadditivity:

$$I(A;B|C) = H(AC) + H(BC) - H(ABC) - H(C) \geq 0$$

Additivity lets us calculate answers

$C($  $\otimes$  $)$

$= C($  $)+C($  $)$

Classical Capacity of Classical Channel

Nonadditivity is the rule
Especially quantumly

Blessing: Better rates for classical and quantum communication.

Curse:
Mostly don't know capacities, distillable entanglement, etc.

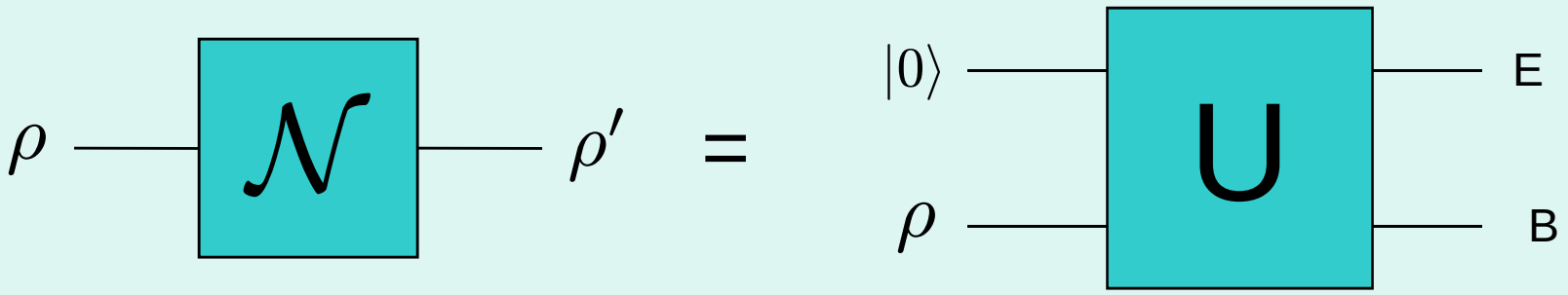
Have upper and lower bounds that are far apart.



Noisy Quantum Channels

- Noiseless quantum evolution: $\rho \rightarrow U\rho U^{\dagger}$

  Unitary satisfies $U^{\dagger}U = I$
- Noisy quantum evolution: unitary interaction with inaccessible environment



$$f(\mathcal{N}) = \mathbf{max}_{\phi_{V_1 \ldots V_n A}} f(\mathcal{N}, \phi_{V_1 \ldots V_n A})$$

Untangling Additivity

$$f(\mathcal{N}) = \mathbf{max}_{\phi_{V_1 \ldots V_n A}} f(\mathcal{N}, \phi_{V_1 \ldots V_n A})$$

$$f(\mathcal{N}, \phi_{V_1 \ldots V_n A}) = \sum_{s \in \mathcal{P}(V_1 \ldots V_n B E)} \alpha_s H(s)$$

Basically all additivity proofs do this:

Take ϕ_{12} that gives a value $f(\mathcal{N} \otimes \mathcal{M}, \phi_{12})$

1) Decoupling: $\phi_{12} \rightarrow \phi_1, \phi_2$

2) Apply strong subadditivity to show:

$$f(\mathcal{N} \otimes \mathcal{M}, \phi_{12}) \leq f(\mathcal{N}, \phi_1) + f(\mathcal{M}, \phi_2)$$

Goal: classify entropic additive formulas of this sort

"uniformly additive"

All uniformly additive entropy formulas

"All the Formulas That's Fit to Print"

One Auxiliary Variable:

case	(a,b)	$\tilde{M}_1$	$\tilde{M}_2$	equivalents	Additive Cone	Extreme Rays
1.	(3,3)	B_1E_1	B_2E_2	(0,0)	$\alpha_V + \alpha_{BV} + \alpha_{EV} \geq 0$ $\alpha_V + \alpha_{BV} \geq 0$ $\alpha_V + \alpha_{EV} \geq 0$ $\alpha_V \geq 0$	$-H(E\|BV)$ $-H(E\|V)$ $-H(B\|EV)$ $-H(B\|V)$
2.	(3,2)	B_1E_1	E_2	$(2,3),(3,1)$ $(1,3),(1,0),(0,1)$ $(2,0),(0,2)$	$\alpha_{BV} \leq 0$ $\alpha_V + \alpha_{BV} \geq 0$	$-H(BE\|V)$ $\pm H(B\|EV)$ $-H(B\|V)$
3.	(3,0)	B_1E_1	ϕ	(0,3)	$\alpha_{EV} \leq 0$ $\alpha_{BV} \leq 0$	$H(E\|BV)$ $-H(E\|V)$ $\pm H(BE\|V)$
4.	(1,1)	B_1	B_2	(2,2)	$\alpha_{EV} = 0$ $\alpha_V \geq 0$ $\alpha_{BEV} \geq 0$	$-H(B\|V)$ $H(E\|BV)$
5.	(1,2)	B_1	E_2	(2,1)	$\alpha_{BEV} \geq 0$ $\alpha_V \geq 0$	$\pm[H(EV) - H(BV)]$ $H(E\|BV)$ $-H(E\|V)$

Rest of talk: explain how to get this table and what it means

Outline

- Standard Additivity Proofs
- Results: all uniformly additive formulas
- Completely coherent information: a new additive quantity
- Further Observations

Canonical example: Entanglement Assisted Capacity

- Entanglement assisted capacity:

$$C_{ea}(\mathcal{N}) = \mathbf{max}_{\phi_{VA}} I(V;B)$$

with I(V;B) = H(V)+H(B)-H(VB).

$$C_{ea}(\mathcal{N} \otimes \mathcal{M}) = C_{ea}(\mathcal{N}) + C_{ea}(\mathcal{M})$$

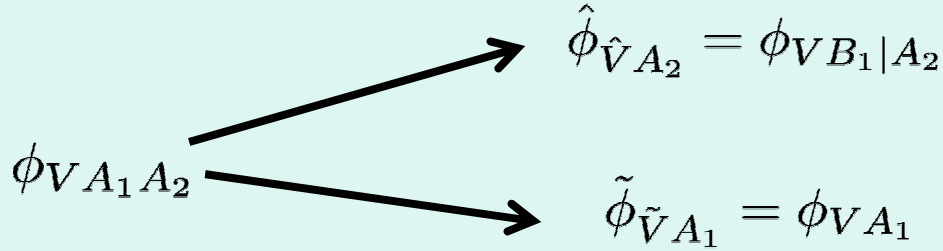
Proof:

Let $\phi_{VA_1A_2}$ be optimal for $C_{ea}(\mathcal{N} \otimes \mathcal{M})$.

$$\begin{aligned} I(V;B_1B_2) &= I(V;B_1) + I(V;B_2|B_1) \\ &= I(V;B_1) + I(VB_1;B_2) - I(B_1;B_2) \\ &\leq I(V;B_1) + I(VB_1;B_2) \leq C_{ea}(\mathcal{N}) + C_{ea}(\mathcal{M}) \end{aligned}$$

Two key steps

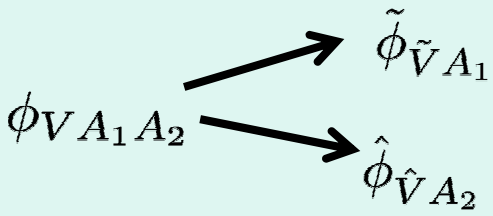
- Decoupling

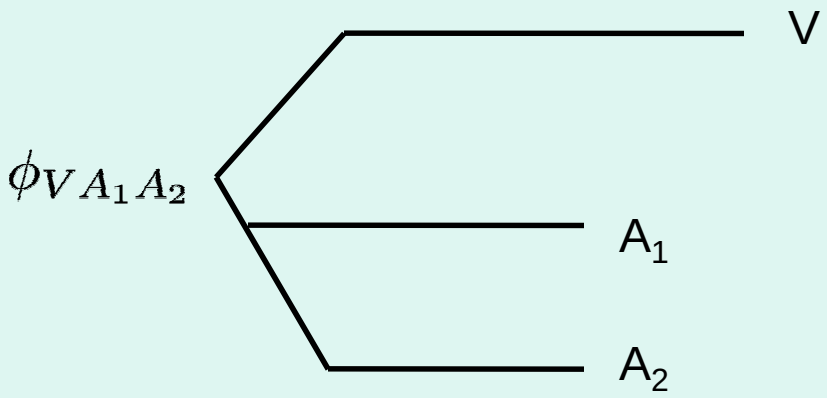


- Entropy inequality

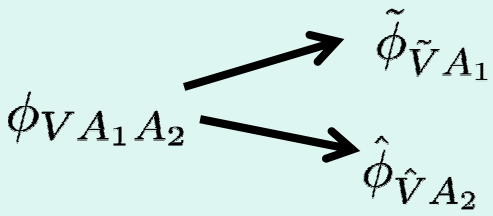
$$\begin{aligned} I(V;B_1B_2) &= I(V;B_1) + I(V;B_2|B_1) \\ &= I(V;B_1) + I(VB_1;B_2) - I(B_1;B_2) \\ &\leq I(V;B_1) + I(VB_1;B_2) \leq C_{ea}(\mathcal{N}) + C_{ea}(\mathcal{M}) \end{aligned}$$

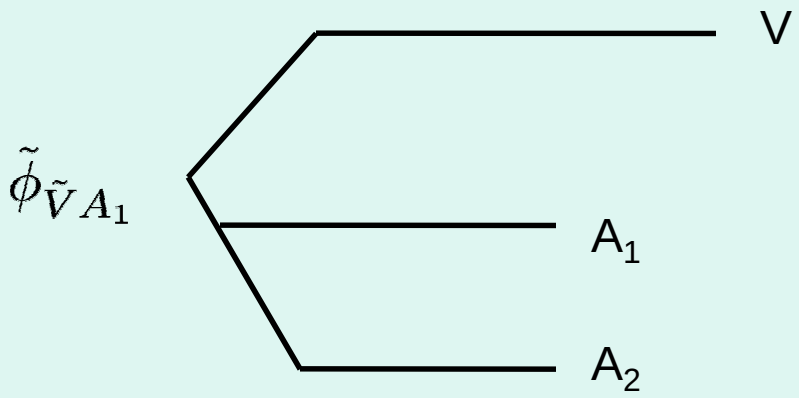
Standard Decouplings





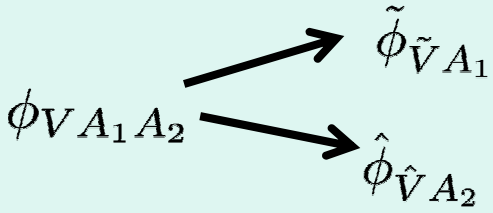
Standard Decouplings

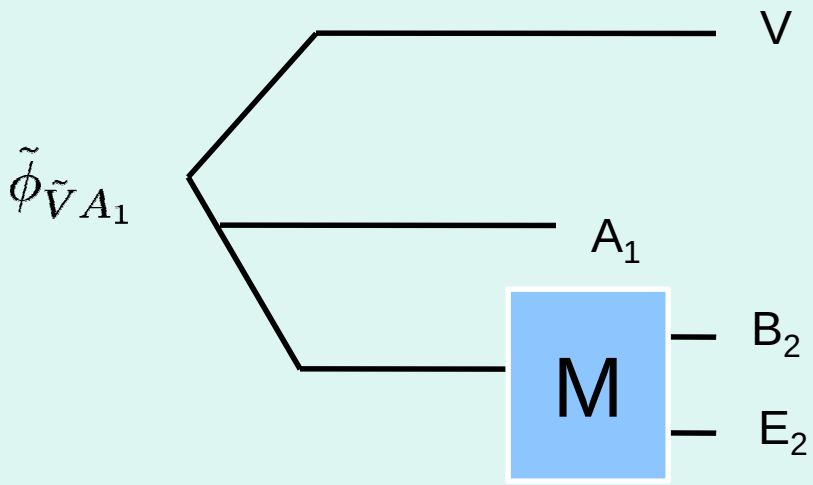




What to do
With A_2?

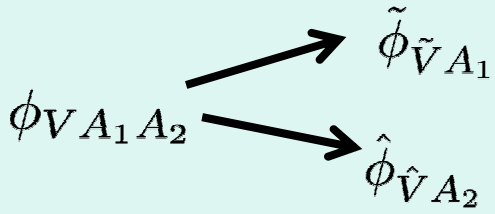
Standard Decouplings

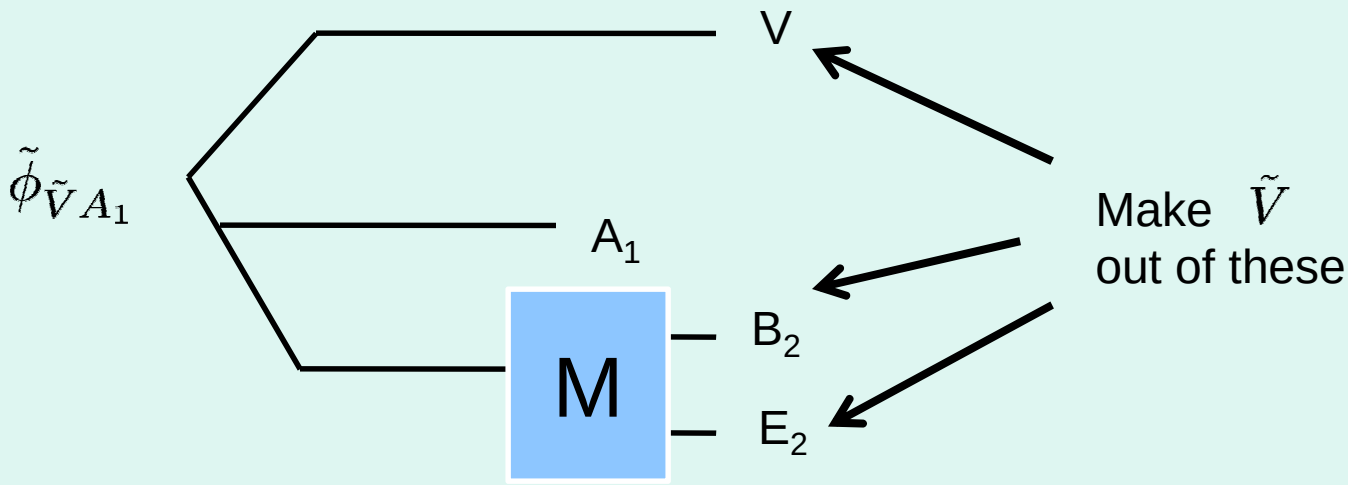




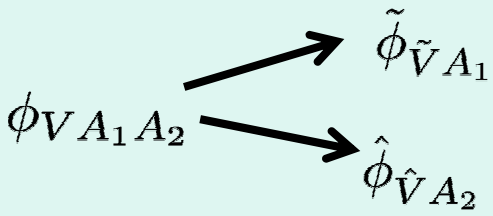
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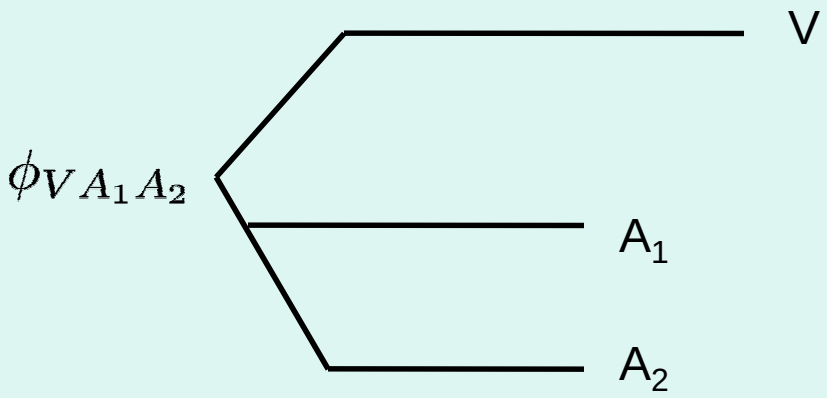
Standard Decouplings



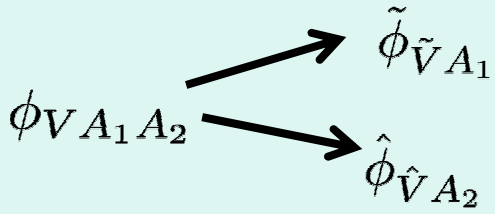


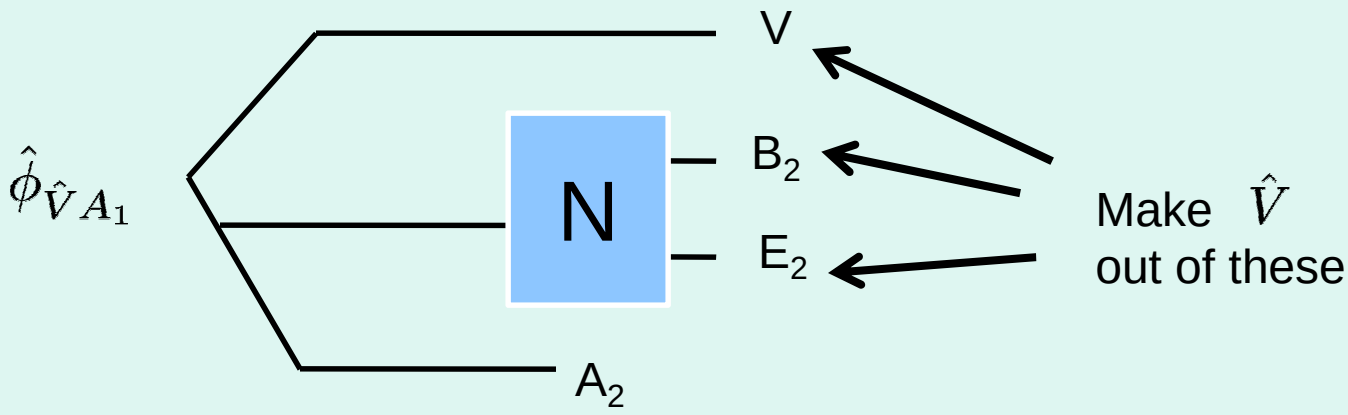
Standard Decouplings





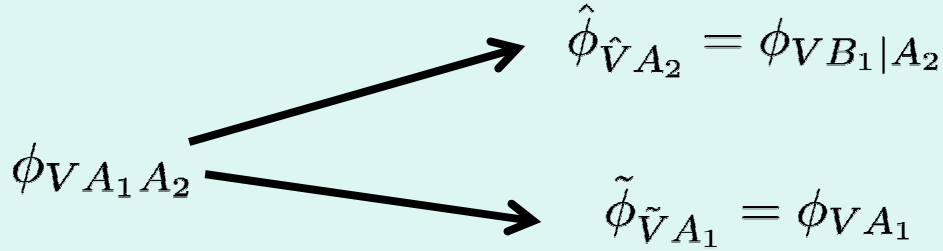
Standard Decouplings





Two key steps

- Decoupling



- Entropy inequality

$$\begin{aligned} I(V;B_1B_2) &= I(V;B_1) + I(V;B_2|B_1) \\ &= I(V;B_1) + I(VB_1;B_2) - I(B_1;B_2) \\ &\leq I(V;B_1) + I(VB_1;B_2) \leq C_{ea}(\mathcal{N}) + C_{ea}(\mathcal{M}) \end{aligned}$$

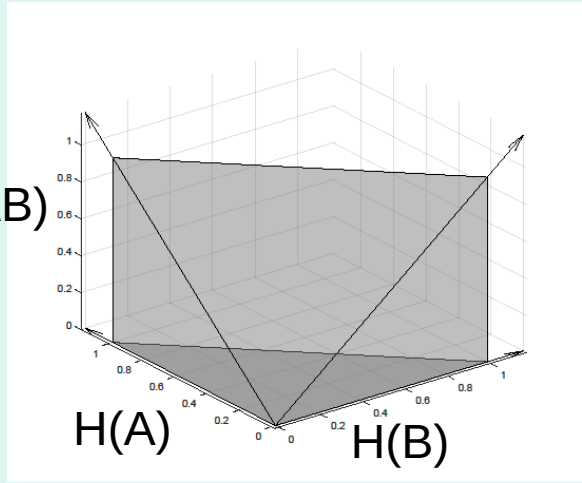
Entropy Inequalities

Strong subadditivity:

$$I(A;B|C) = H(AC) + H(BC) - H(ABC) - H(C) \geq 0$$

Also: $H(A|C) + H(A|D) \geq 0$

There may be more,
but we don't know them.



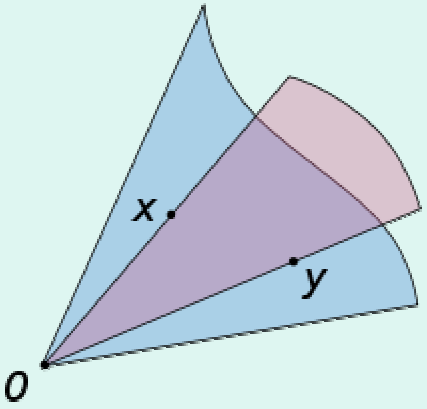
Entropy inequalities

- For n systems A_1 … A_n, there are 2^n -1 different entropies:

$$(H(A_1), ..., H(A_n), H(A_1A_2)..., H(A_1...A_n))$$

- Not all vectors can be realized. Realizable vectors form a cone
- Cone bounded by various planes defined by strong subadditivity
- Can test if an identity is satisfied by Linear Programming.
- There are other “non-Shannon”classical inequalities, but for quantum we don’t know (though there probably are).





Outline

- ~~Standard Additivity Proofs~~
- Results: all uniformly additive formulas
- Completely coherent information: a new additive quantity
- Further Observations

Entropic formulas

$$f_\alpha(\mathcal{N}, \phi_{V_1 \ldots V_n A}) = \sum_{s \in \mathcal{P}(V_1 \ldots V_n BE)} \alpha_s H(s)$$

Fix a standard decoupling $\phi_{V A_1 A_2} \to (\tilde{\phi}_{\tilde{V} A_1}, \hat{\phi}_{\hat{V} A_2})$

$$\Delta\left(f_\alpha, U_{\mathcal{N}_1}, U_{\mathcal{N}_2}, \phi_{V_1 \ldots V_n A_1 A_2}, \tilde{\phi}_{\tilde{V}_1 \ldots \tilde{V}_n A_1}, \hat{\phi}_{\hat{V}_1 \ldots \hat{V}_n A_2}\right) =$$

$$f_\alpha(U_{\mathcal{N}_1}, \tilde{\phi}_{\tilde{V}_1 \ldots \tilde{V}_n A_1}) + f_\alpha(U_{\mathcal{N}_2}, \hat{\phi}_{\hat{V}_1 \ldots \hat{V}_n A_2}) - f_\alpha(U_{\mathcal{N}_1} \otimes U_{\mathcal{N}_2}, \phi_{V_1 \ldots V_n A_1 A_2})$$

Find α such that $\Delta(f_\alpha) \geq 0$ for all $\mathcal{N}_1$ $\mathcal{N}_2$, $\phi_{V_1 \ldots V_n A_1 A_2}$.



Zero Auxiliary Variables

$$f_\alpha(\mathcal{N}, \phi_A) = \alpha_B H(B) + \alpha_E H(E) + \alpha_{BE} H(BE)$$

Decoupling: $\phi_{A_1 A_2} \to (\phi_{A_1}, \phi_{A_2})$

$$\begin{aligned}\Delta(\alpha, \phi_{A_1 A_2}) &= f_\alpha(\mathcal{N}_1, \phi_{A_1}) + f_\alpha(\mathcal{N}_2, \phi_{A_2}) - f_\alpha(\mathcal{N}_2, \phi_{A_1 A_2}) \\ &= \alpha_B I(B_1; B_2) + \alpha_E I(E_1; E_2) + \alpha_{BE} I(B_1 E_1; B_2 E_2)\end{aligned}$$

When is $\Delta(\alpha, \phi_{A_1 A_2}) \geq 0$?

Zero Auxiliary Variables

$$\text{When is } \Delta(\alpha, \phi_{A_1 A_2}) \geq 0?$$

Rays

$$f_\alpha = \lambda_1 H(B) + \lambda_2 H(E) + \lambda_3 H(B|E) + \lambda_4 H(E|B)$$

$$\lambda_i \geq 0$$

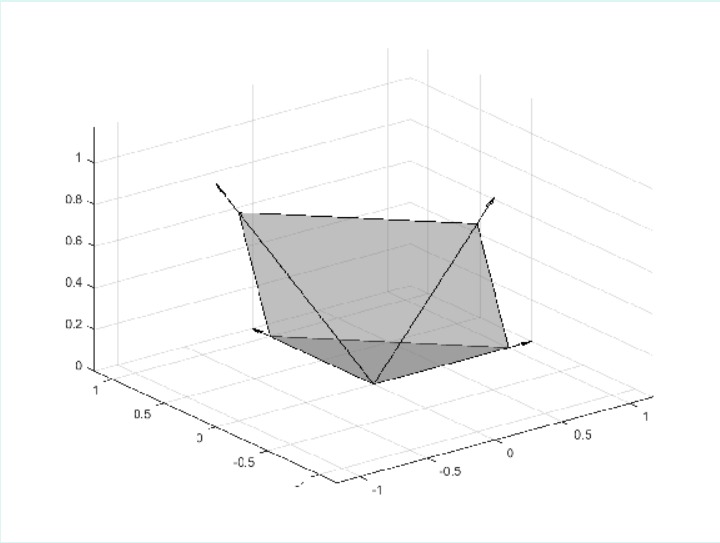
Faces

$$\alpha_B + \alpha_{BE} \geq 0$$

$$\alpha_E + \alpha_{BE} \geq 0$$

$$\alpha_B + \alpha_E + \alpha_{BE} \geq 0$$

$$\alpha_{BE} \geq 0.$$



Anything inside the cone is uniformly additive.

Outside the cone, there is A state that makes $\Delta < 0$

To show, find state that makes, e.g.,

$$\Delta = \alpha_B + \alpha_{BE}$$

One Auxiliary Variable

Enough to consider

$$f_\alpha^V(\mathcal{N}, \phi_{VA}) = \alpha_V H(V) + \alpha_{BV} H(BV) + \alpha_{EV} H(EV) + \alpha_{BEV} H(BEV)$$

Fix a standard decoupling:

$\tilde{V} \in \{V, B_2V, E_2V, B_2E_2V\}$ and

$\hat{V} \in \{V, B_1V, E_1V, B_1E_1V\}$

These are labeled by (a, b) $a, b = 0...3$

Define $\Delta_{(a,b)}(\alpha^V)$, find $\alpha^V = (\alpha_V, \alpha_{BV}, \alpha_{EV}, \alpha_{BEV})$ with $\Delta_{(a,b)} \geq 0$.

This gives a cone of additive functions for each (a,b)

One Auxiliary Variable

case	(a,b)	$\tilde{M}_1$	$\tilde{M}_2$	equivalents	Additive Cone	Extreme Rays
1.	(3,3)	B_1E_1	B_2E_2	(0,0)	$\alpha_V + \alpha_{BV} + \alpha_{EV} \geq 0$ $\alpha_V + \alpha_{BV} \geq 0$ $\alpha_V + \alpha_{EV} \geq 0$ $\alpha_V \geq 0$	$-H(E\|BV)$ $-H(E\|V)$ $-H(B\|EV)$ $-H(B\|V)$
2.	(3,2)	B_1E_1	E_2	$(2,3),(3,1)$ $(1,3),(1,0),(0,1)$ $(2,0),(0,2)$	$\alpha_{BV} \leq 0$ $\alpha_V + \alpha_{BV} \geq 0$	$-H(BE\|V)$ $\pm H(B\|EV)$ $-H(B\|V)$
3.	(3,0)	B_1E_1	ϕ	(0,3)	$\alpha_{EV} \leq 0$ $\alpha_{BV} \leq 0$	$H(E\|BV)$ $-H(E\|V)$ $\pm H(BE\|V)$
4.	(1,1)	B_1	B_2	(2,2)	$\alpha_{EV} = 0$ $\alpha_V \geq 0$ $\alpha_{BEV} \geq 0$	$-H(B\|V)$ $H(E\|BV)$
5.	(1,2)	B_1	E_2	(2,1)	$\alpha_{BEV} \geq 0$ $\alpha_V \geq 0$	$\pm[H(EV) - H(BV)]$ $H(E\|BV)$ $-H(E\|V)$

$$f_\alpha^V(\mathcal{N}, \phi_{VA}) = \alpha_V H(V) + \alpha_{BV} H(BV) + \alpha_{EV} H(EV) + \alpha_{BEV} H(BEV)$$

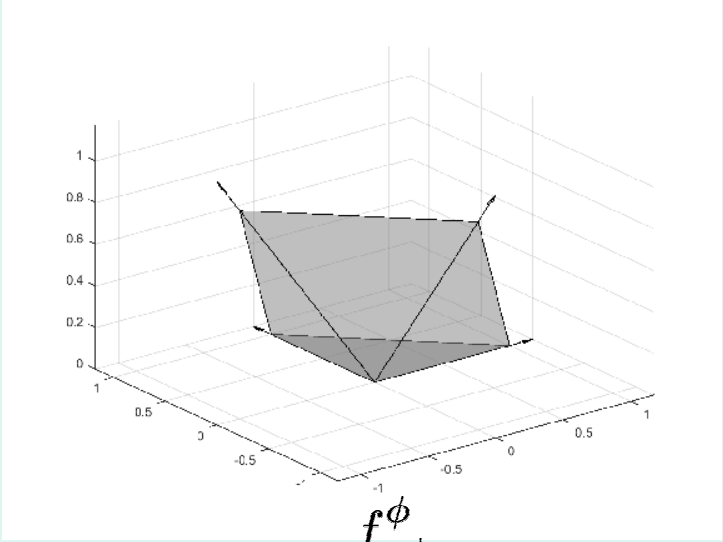
One Auxiliary Variable

Most general: $f_\alpha = f^\phi_\alpha + f^V_{\alpha^V}$

With

$$f^\phi_\alpha = \lambda_1 H(B) + \lambda_2 H(E) + \lambda_3 H(B|E) + \lambda_4 H(E|B)$$

$$f^V_\alpha(\mathcal{N}, \phi_{VA}) = \alpha_V H(V) + \alpha_{BV} H(BV) + \alpha_{EV} H(EV) + \alpha_{BEV} H(BEV)$$



$f^\phi_{\alpha^\phi}$

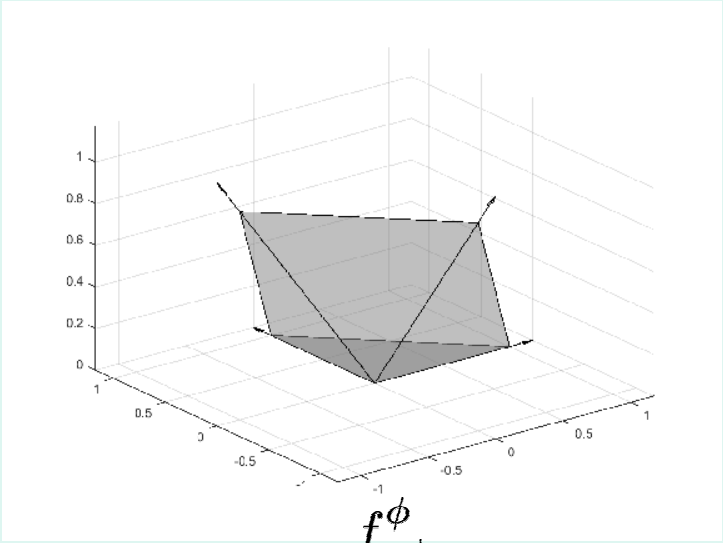
case	(a,b)	M_1	M_2	equivalents	Additive Cone	Extreme Rays
1.	(3,3)	B_1E_1	B_2E_2	(0,0)	$\alpha_V + \alpha_{BV} + \alpha_{EV} \geq 0$ $\alpha_V + \alpha_{BV} \geq 0$ $\alpha_V + \alpha_{EV} \geq 0$ $\alpha_V \geq 0$	$-H(E\|BV)$ $-H(E\|V)$ $-H(B\|EV)$ $-H(B\|V)$
2.	(3,2)	B_1E_1	E_2	$(2,3),(3,1)$ $(1,3),(1,0),(0,1)$ $(2,0),(0,2)$	$\alpha_{BV} \leq 0$ $\alpha_V + \alpha_{BV} \geq 0$	$-H(BE\|V)$ $\pm H(B\|EV)$ $-H(B\|V)$
3.	(3,0)	B_1E_1	ϕ	(0,3)	$\alpha_{EV} \leq 0$ $\alpha_{BV} \leq 0$	$H(E\|BV)$ $-H(E\|V)$ $\pm H(BE\|V)$
4.	(1,1)	B_i	B_2	(2,2)	$\alpha_{EV} = 0$ $\alpha_V \geq 0$ $\alpha_{BEV} \geq 0$	$-H(B\|V)$ $H(E\|BV)$
5.	(1,2)	B_i	E_2	(2,1)	$\alpha_{BEV} \geq 0$ $\alpha_V \geq 0$	$\pm[H(EV) - H(BV)]$ $H(E\|BV)$ $-H(E\|V)$

$f^V_{\alpha^V}$

Many Auxiliary Variables

$$\text{Most general: } f_\alpha = f_\alpha^\phi + f_{\alpha V_1}^{V_1} + f_{\alpha V_2}^{V_2} + f_{\alpha V_1 V_2}^{V_1 V_2}$$

Just pick each individual function from the appropriate row in the table.



$f_{\alpha^\phi}^{\phi}$

case	(a,b)	M_1	M_2	equivalents	Additive Cone	Extreme Rays
1.	(3,3)	B_1E_1	B_2E_2	(0,0)	$\alpha_V + \alpha_{BV} + \alpha_{EV} \geq 0$ $\alpha_V + \alpha_{BV} \geq 0$ $\alpha_V + \alpha_{EV} \geq 0$ $\alpha_V \geq 0$	$-H(E\|BV)$ $-H(E\|V)$ $-H(B\|EV)$ $-H(B\|V)$
2.	(3,2)	B_1E_1	E_2	(2,3), (3,1) (1,3), (1,0), (0,1) (2,0), (0,2)	$\alpha_{BV} \leq 0$ $\alpha_V + \alpha_{BV} \geq 0$	$-H(BE\|V)$ $\pm H(B\|EV)$ $-H(B\|V)$
3.	(3,0)	B_1E_1	ϕ	(0,3)	$\alpha_{EV} \leq 0$ $\alpha_{BV} \leq 0$	$H(E\|BV)$ $-H(E\|V)$ $\pm H(BE\|V)$
4.	(1,1)	B_i	B_2	(2,2)	$\alpha_{EV} = 0$ $\alpha_V \geq 0$ $\alpha_{BEV} \geq 0$	$-H(B\|V)$ $H(E\|BV)$
5.	(1,2)	B_i	E_2	(2,1)	$\alpha_{BEV} \geq 0$ $\alpha_V \geq 0$	$\pm[H(EV) - H(BV)]$ $H(E\|BV)$ $-H(E\|V)$

$f_{\alpha^V}^{V}$

Outline

- ~~Standard Additivity Proofs~~
- ~~Results: all uniformly additive formulas~~
- Completely coherent information: a new additive quantity
- Further Observations

Completely Coherent Information

case	(a,b)	$\tilde{M}_1$	$\tilde{M}_2$	equivalents	Additive Cone	Extreme Rays
1.	(3,3)	B_1E_1	B_2E_2	(0,0)	$\alpha_V + \alpha_{BV} + \alpha_{EV} \geq 0$ $\alpha_V + \alpha_{BV} \geq 0$ $\alpha_V + \alpha_{EV} \geq 0$ $\alpha_V \geq 0$	$-H(E\|BV)$ $-H(E\|V)$ $-H(B\|EV)$ $-H(B\|V)$
2.	(3,2)	B_1E_1	E_2	$(2,3),(3,1)$ $(1,3),(1,0),(0,1)$ $(2,0),(0,2)$	$\alpha_{BV} \leq 0$ $\alpha_V + \alpha_{BV} \geq 0$	$-H(BE\|V)$ $\pm H(B\|EV)$ $-H(B\|V)$
3.	(3,0)	B_1E_1	ϕ	(0,3)	$\alpha_{EV} \leq 0$ $\alpha_{BV} \leq 0$	$H(E\|BV)$ $-H(E\|V)$ $\pm H(BE\|V)$
4.	(1,1)	B_1	B_2	(2,2)	$\alpha_{EV} = 0$ $\alpha_V \geq 0$ $\alpha_{BEV} \geq 0$	$-H(B\|V)$ $H(E\|BV)$
5.	(1,2)	B_1	E_2	(2,1)	$\alpha_{BEV} \geq 0$ $\alpha_V \geq 0$	$\pm[H(EV) - H(BV)]$ $H(E\|BV)$ $-H(E\|V)$

Completely Coherent Information

case	(a,b)	$\tilde{M}_1$	$\tilde{M}_2$	equivalents	Additive Cone	Extreme Rays
1.	(3,3)	B_1E_1	B_2E_2	(0,0)	$\alpha_V + \alpha_{BV} + \alpha_{EV} \geq 0$ $\alpha_V + \alpha_{BV} \geq 0$ $\alpha_V + \alpha_{EV} \geq 0$ $\alpha_V \geq 0$	$-H(E\|BV)$ $-H(E\|V)$ $-H(B\|EV)$ $-H(B\|V)$
2.	(3,2)	B_1E_1	E_2	$(2,3),(3,1)$ $(1,3),(1,0),(0,1)$ $(2,0),(0,2)$	$\alpha_{BV} \leq 0$ $\alpha_V + \alpha_{BV} \geq 0$	$-H(BE\|V)$ $\pm H(B\|EV)$ $-H(B\|V)$
3.	(3,0)	B_1E_1	ϕ	(0,3)	$\alpha_{EV} \leq 0$ $\alpha_{BV} \leq 0$	$H(E\|BV)$ $-H(E\|V)$ $\pm H(BE\|V)$
4.	(1,1)	B_1	B_2	(2,2)	$\alpha_{EV} = 0$ $\alpha_V \geq 0$ $\alpha_{BEV} \geq 0$	$-H(B\|V)$ $H(E\|BV)$
5.	(1,2)	B_1	E_2	(2,1)	$\alpha_{BEV} \geq 0$ $\alpha_V \geq 0$	$\pm[H(EV) - H(BV)]$ $H(E\|BV)$ $-H(E\|V)$

Completely Coherent Information: Properties

$$I^{cc}(\mathcal{N}) = \max_{\phi_{VA}} [H(VB) - H(VE)]$$

- Symmetric in B ←→ E
- Lower bound for cost of swapping B and E.
- Upper bound for simultaneous quantum communication rate to B and E
- For degradable channels, $I^{cc}(N) = Q(N) = Q^{(1)}(N)$
- WANT: cardinality bound on V

Generalizing Completely Coherent Information

We can replace the entropies in I^{cc} with any function $J(\rho)$ that satisfies:

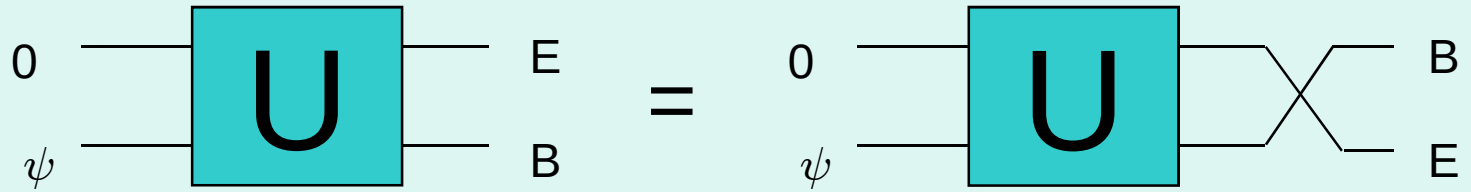
- $J(\rho_{V1B_1} \otimes \rho_{V2B_2}) = J(\rho_{V1B_1}) + J(\rho_{V2B_2})$
- $F_J(N) = \max [J(VB) - J(VE)]$

Furthermore, if J is monotonic under CP maps, $F_J(N)$ doesn't need auxiliary variable

J could be: renyi entropy, sandwiched entropy, (α, z)-entropies…

Connection to Symmetric Side Channels

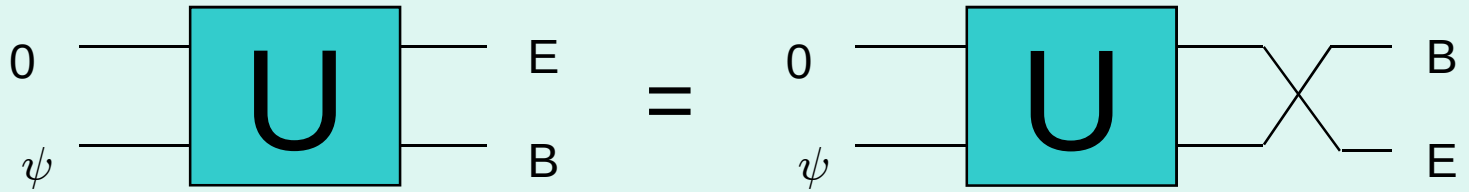
Quantum capacity with symmetric zero-capacity channel
Additive upper bound for Q



$$
\begin{aligned}
Q_{ss}(\mathcal{N}) &= (1/2)[I(V_1;B|V_2) - I(V_1;E|V_2)] \\
&= (1/2)[H(BV_2) - H(BV_1V_2) - H(EV_2) + H(EV_1V_2)] \\
&= (1/2)[H(BV_2) - H(EV_2) + H(EV_1V_2) - H(BV_1V_2)]
\end{aligned}
$$

Connection to Symmetric Side Channels

Quantum capacity with symmetric zero-capacity channel
Additive upper bound for Q



$$
\begin{aligned}
Q_{ss}(\mathcal{N}) &= (1/2)[I(V_1;B|V_2) - I(V_1;E|V_2)] \\
&= (1/2)[H(BV_2) - H(BV_1V_2) - H(EV_2) + H(EV_1V_2)] \\
&= (1/2)[\underbrace{H(BV_2) - H(EV_2)}_{I^{cc}} + \underbrace{H(EV_1V_2) - H(BV_1V_2)}_{I^{cc}}]
\end{aligned}
$$

Outline

- ~~Standard Additivity Proofs~~
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A Classical-Quantum Coincidence

- You can do this whole game for classical entropic formulas too.
- You get *exactly* the same set of uniformly additive functions.
- Could have been more, since there are more classical inequalities: $H(X|Y) \geq 0$
- But uniform additivity only uses strong subadditivity.

Further Directions

- Classically, there are additive functions that are not uniformly additive (e.g. H_{min})
- Single-letter entropic constraints that imply additivity of coherent information (informationally degradable). Only one?
- Applies to rate regions of multi-user information theory
- Constraint on states optimized over

Summary



- Additivity simplifies, but is rare
- Typical additivity proof has two steps:
  1) Decoupling 2) Apply entropy inequalities
- Uniform additivity: standard decoupling + entropy inequalities
- Classified all uniformly additive formulas
- Completely coherent information

Open Questions

- Constrained additivity: 1) constraints on channels 2) constraints on states
- Completely coherent information: operational meaning, cardinality bound
- Understand classical-quantum correspondence better. Coincidence?
- Apply to formulas with c-q states (χ, P)? Decoupling is a challenge there.

THANK YOU